process in a complex organism is regulated by one or more hormones: maintenance of blood pressure, blood volume and electrolyte balance, embryogenesis, hunger, eating behaviour, digestion - to name but a few. Endocrine glands, which are special groups of cells, make hormones. The major endocrine glands are the pituitary, pineal, thymus, thyroid, adrenal glands, and pancreas. In addition, men produce hormones in their testes and women produce them in their ovary. Chemically, hormones may be classified as either protein (e.g. insulin, epinephrine) or steroids (e.g. estrogen, androgen).Hormones are classified according to the distance over which they act as, endocrine, paracrine and autocrine hormones

Endocrine hormones act on cells distant from the site of their release. Example: insulin and epinephrine are synthesized and released in the bloodstream by specialized ductless endocrine glands.

Paracrine hormones (alternatively, local mediators) act only on cells close to the cell that released them. For example, interleukin-1 (IL-1)

Autocrine hormones act on the same cell that released them. For example, protein growth factor interleukin-2 (IL-2).

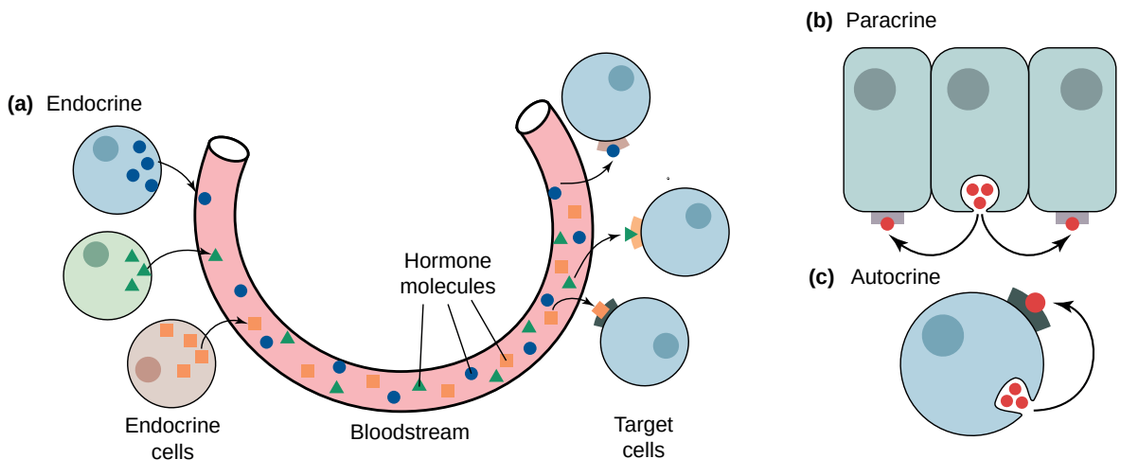

Figure 14.22 Endocrine Paracrine and Autocrine hormones

Only those cells with a specific receptor for a given hormone will respond to its presence even though nearly all cells in the body may be exposed to the hormone. Hormonal messages are therefore quite specifically addressed.

XII U14-Biomolecules_New.indd 267 11-01-2021 16:38:04